

Generation Interconnection System Impact Study Report

For Queue Project Q21-1863

Uintah 345 kV
February 25, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1863. The Interconnection Customer proposes a 275.6 MW solar photovoltaic facility interconnecting at the Uintah 345 kV substation in the Virginia area. The request was received on August 19, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 552 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$53,236,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Untah-Leesburg 138kV line	Claytonia Substation-Uintah 345kV	219.5	231	88.4
Trumann-Rossmoor South 138kV	G-P Gypsum-Brighton 138kV	177.1	245	5.0
Tramway-Rossmoor South 138kV	G-P Gypsum-Brighton 138kV	157.5	313	5.0
Tramway-Bondurant 138kV Line 2	Bondurant-Tramway 138kV Circuit 1	148.5	376	5.8
Montgomery Village-Bondurant 138kV Cir. 2	Montgomery Village-Bondurant 138kV	146.1	396	5.8
Bondurant-Tramway 138kV #1	Bondurant-Tramway 138kV Cir. 2	143.0	391	5.8
Montgomery Village-Bondurant 138kV	Bondurant-Montgomery Village 138kV #2	133.7	433	5.8
Carlotta-Ritzville 138kV L	Air Force Base 345/138 kV TX	128.3	276	10.9
Monroe Ne-South Meadow 345kV line	Channel Energy Ctr-Monroe Ne 500kV	111.3	1,187	6.0
Tiger Bay 345/138kV Autotransformer	Air Force Base 345/138 kV Transformer Bank	103.4	549	5.2
Sterling Park Substation-Leesburg 138kV	Uintah Substation-Claytonia 345kV	101.7	310	46.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Utah-Leesburg 138kV line	Rebuild line with higher-capacity conductor	\$2,687,000	63% (\$1,687,000)
Trumann-Rossmoor South 138kV	Rebuild line with higher-capacity conductor	\$8,447,000	100% (\$8,447,000)
Tramway-Rossmoor S 138kV	Rebuild line with higher-capacity conductor	\$1,607,000	100% (\$1,607,000)
Bondurant Substation-Tramway 138kV	Rebuild line with higher-capacity conductor	\$1,761,000	100% (\$1,761,000)
Montgomery Village-Bondurant 138kV Cir. 2	Rebuild line with higher-capacity conductor	\$7,314,000	100% (\$7,314,000)
Bondurant-Tramway 138kV #1	Rebuild line with higher-capacity conductor	\$1,453,000	100% (\$1,453,000)
Montgomery Village-Bondurant 138kV	Rebuild line with higher-capacity conductor	\$6,249,000	100% (\$6,249,000)
Ritzville-Carlotta 138kV	Rebuild line with higher-capacity conductor	\$3,113,000	74% (\$2,287,000)
Monroe Ne-South Meadow 345kV line	Rebuild line with higher-capacity conductor	\$17,550,000	3% (\$466,000)
Tiger Bay 345/138kV Autotransformer	Replace transformer with higher-rated unit	\$19,286,000	100% (\$19,286,000)
Sterling Park Substation-Leesburg 138kV	Reconductor line	\$4,376,000	61% (\$2,679,000)
	Total allocated to project		\$53,236,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.